

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Cache Organization

Lecture No.- 3

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Recap of Previous Lecture



Topic

Cache Memory

Topic

Average Memory Access Time

Topic

Cache Write

Topics to be Covered



Topic

Cache Write

Topic

Cache Mapping

Topic

Direct Mapping



Topic : T_{avg} in Write Back Cache



$$T_{avg} = H * t_{cm} + (1-H) (t_{cm} + t_{bt} + x * t_{bt})$$

x = fraction of replaced blocks from cache which are dirty

#Q. The memory access time is 1 nanosecond for a read operation with a hit in cache, 5 nanoseconds for a read operation with a miss in cache, 2 nanoseconds for a write operation with a hit in cache and 10 nanoseconds for a write operation with a miss in cache. Execution of a sequence of instructions involves 100 instruction fetch operations; 60 memory operand read operations and 40 memory operand write operations. The cache hit-ratio is 0.9. The average memory access time (in nanoseconds) in executing the sequence of instructions is?



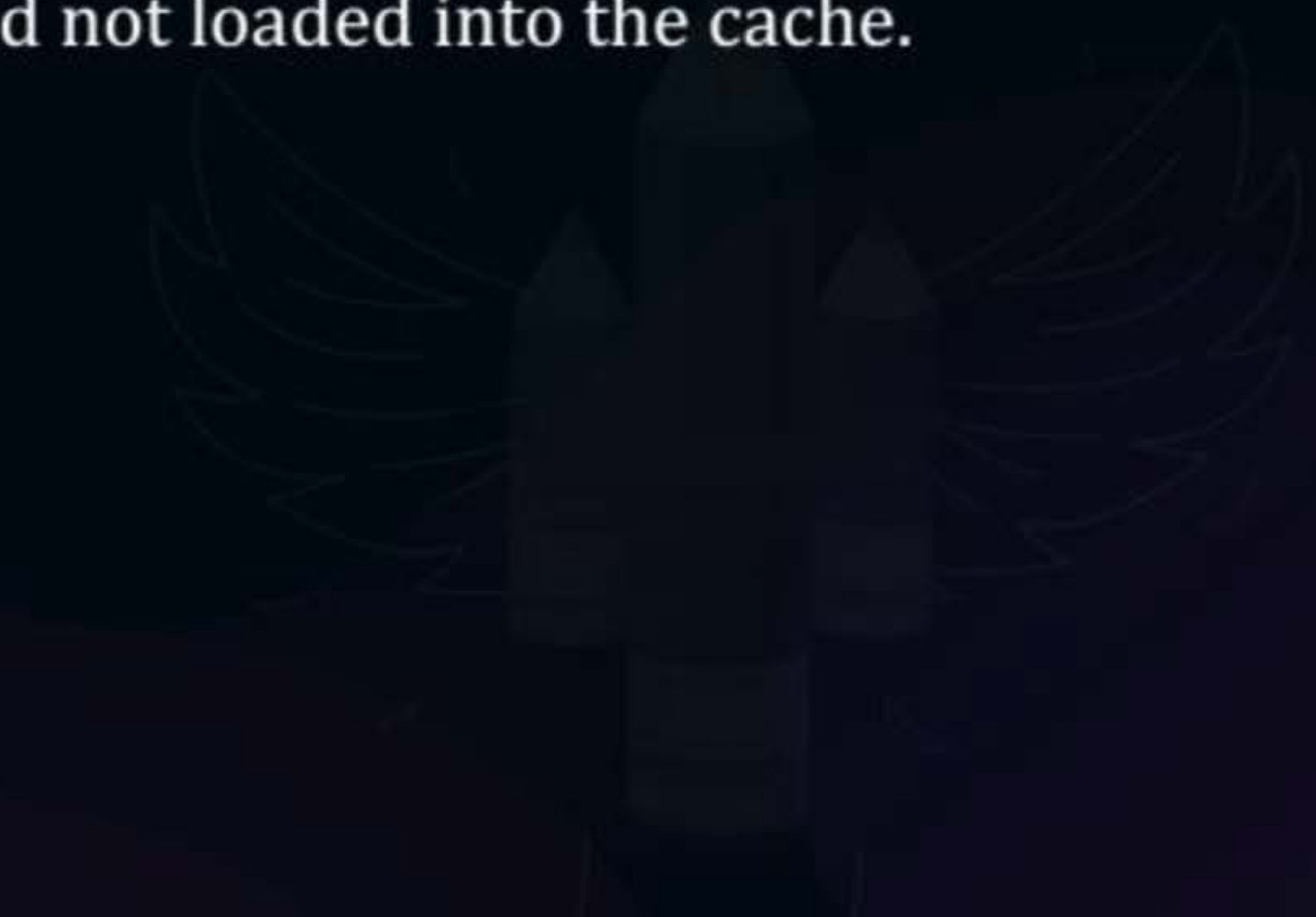
Topic : Write Allocate vs No Write Allocate

Write Allocate: → used with write back cache

The missed block is loaded in cache on a write miss.

No Write Allocate: → used with write through cache

The block is modified in the main memory and not loaded into the cache.





Topic : Write Through with no write allocate

Read

Hit

CPU reads required content from cache

miss

CPU reads required content from mm. CPU brings missed block from mm to cache. If any block is replaced from cache then directly just replace it (no write back needed)

write

Hit

CPU performs write in cache and mm simultaneously

miss

CPU performs write in mm and does not bring the missed block to cache



Topic : Write Back with write allocate

Read

Hit

CPU reads required content from cache

Miss

CPU reads required content from mm.
CPU brings missed block from mm to cache.
If any block is replaced from cache then write it back to mm if it is dirty; otherwise replace it directly.

Write

Hit

CPU performs write in cache; and makes this block dirty.

Miss

CPU first brings missed block from mm to cache then performs write into cache.
and makes this block dirty.
If any block got replaced from cache then write it back to mm if it is dirty.

H.W.

#Q. Size of data sent to main memory from CPU:

1. For write hit, when a write through cache is used?
2. For write miss, when a write through cache is used?
3. For write hit, when a write back cache is used?
4. For write miss, when a write back cache is used?

H.W.

#Q. Size of data sent from main memory to cache:

1. For write hit, when a write through cache is used?
2. For write miss, when a write through cache is used?
3. For write hit, when a write back cache is used?
4. For write miss, when a write back cache is used?

#Q. Consider a system with average cycles per instruction (CPI) is 5 (without any operand access from memory). Consider the average memory access time is 5 cycles and 30% instructions only need 1 memory access for operand then the average CPI is 6.5 ?

$$= 5 + (0.3 * 5)$$

$$= 5 + 1.5$$

$$= 6.5$$

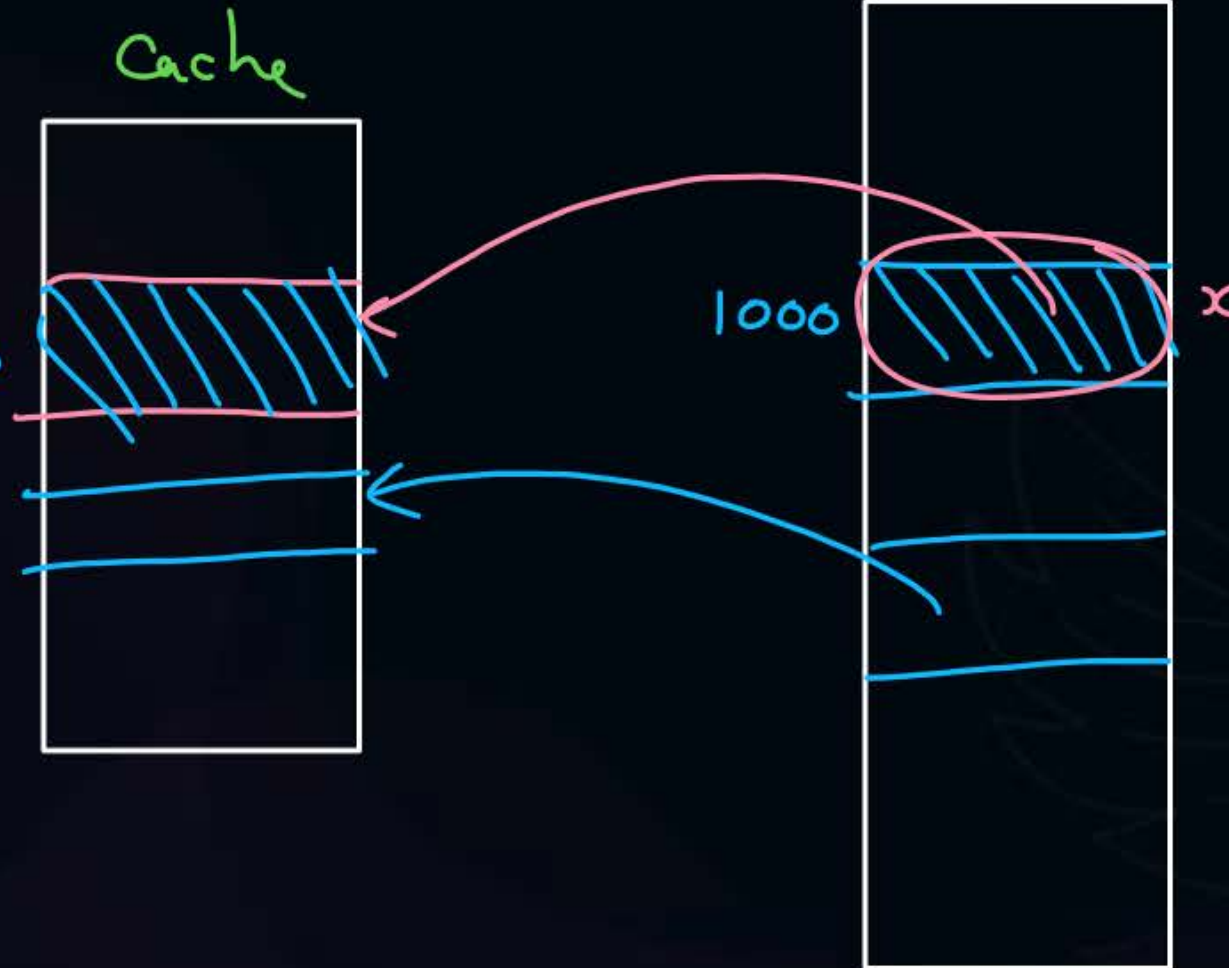


Topic : Cache Mapping

address
CPU generates m always to access mem. system.

cache location
= mapping function (1000)

function(1000)



There is a predefined method or function with the help of which content of mm from a specific address is copied into cache at a specific location.

This predefined method is called as cache mapping.

⇒ mapping helps in searching cache with mm address.

⇒ mapping applied on blocks because the block is copied from mm to cache.



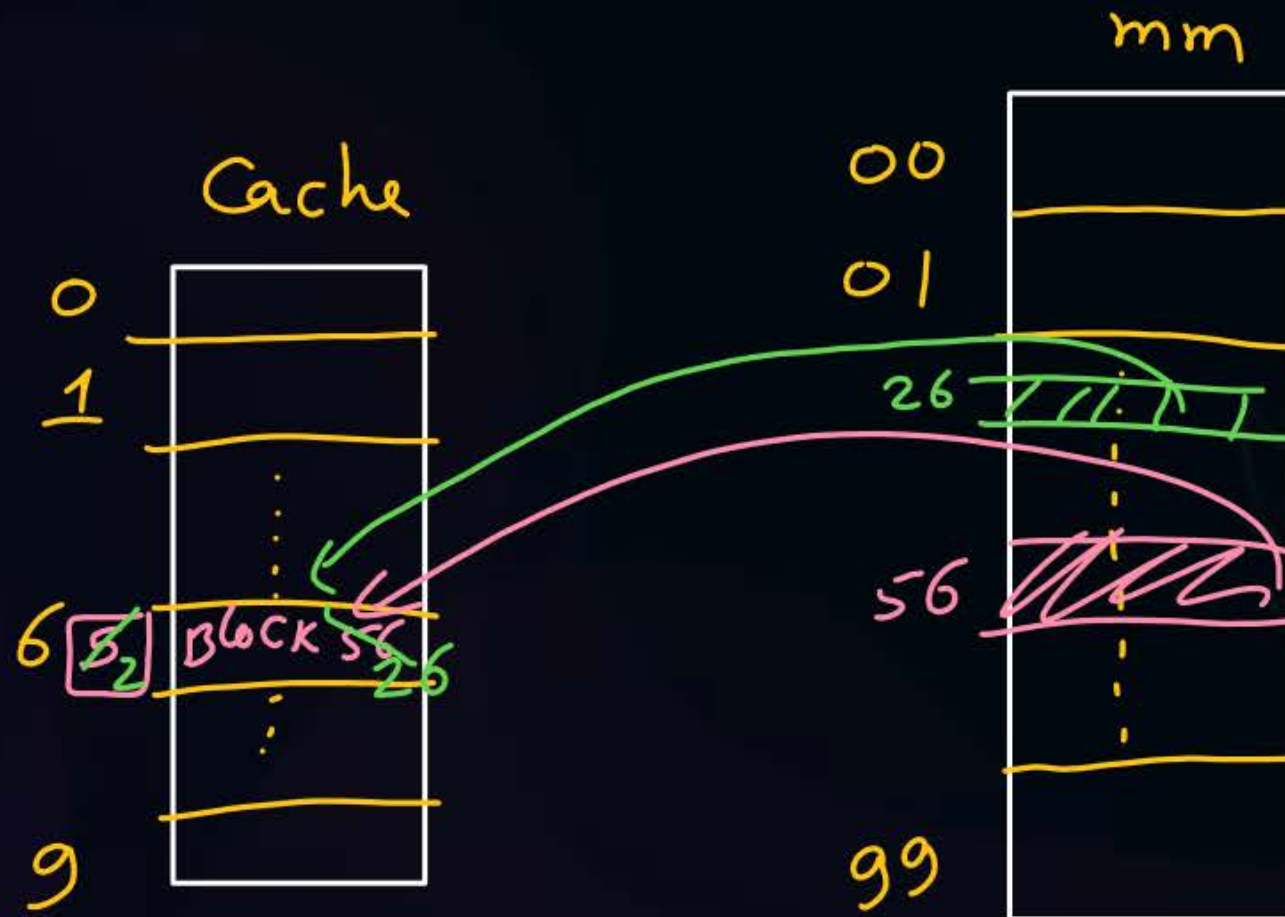
Topic : Cache Mapping

- Direct Mapping
- Set Associative Mapping
- Fully Associative Mapping



Topic : Direct Mapping

- Blocks in cache = 10 (0-9)
- Blocks in Main memory = 100 (00-99)





Topic : Direct Mapping

Block 26

~~Block 56~~

Cache

0 1 2 3 4 5 6 7 8 9

Main
Memory

00	01	02	03	04	05	06	07	08	09
10	11	12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27	28	29
30	31	32	33	34	35	36	37	38	39
:	:	:	:	:	:	:	:	:	:
90	91	92	93	94	95	96	97	98	99



Topic : Direct Mapping



$$\text{cm block no.} = (\text{mm block no.}) \% (\text{no. of blocks in cache})$$

$$\text{Tag} = \left\lfloor \frac{\text{mm block no.}}{\text{no. of blocks in cache}} \right\rfloor$$

Tag $\Rightarrow$ stored in cache along with block to identify which mm block is present in cache currently.



Topic : Direct Mapping

CPU Request (MM block)	Mapping (CM block no.)	Hit /Miss	Comments
56	$56 \% 10 = 6$ CPU searches cache at block 6. No any block in cache at block 6.	miss	CPU brings content of mm block no. 56 at cache block no. 6.
26	$26 \% 10 = 6$ CPU searches cache at block 6.	miss	CPU brings Content of mm block no. 26 at cache block no. 6. by replacing content of mm block no. 56.



Topic : Direct Mapping

	Tag	blocks
0		
1		
2		
	⋮	⋮
6	5/2	Block 56 Block 26
	⋮	⋮
9		

mm block no.

Tag	cm block no.
-----	--------------

ex:-

mm block no. = 56

5	6
---	---

mm block no. = 26

2	6
---	---



Topic : Direct Mapping

cpu request
mm block no.

56

⇒ bring block 56 at 6 with tag 5

26

⇒ $26 \div 10 = 6 \rightarrow$ goto cache at block 6
and compares tag.

Tag = 2

block present in cache has tag = 5 }
2 }

not matching

↓
MISS



2 mins Summary



Topic

Cache Write

Topic

Cache Mapping

Topic

Direct Mapping



Happy Learning

THANK - YOU

